EE-482: Network Security and Cryptography

LECTURES NO 7: *DES Variations*

Engr. Tehseen Ahsan
Assistant Professor, Computer Engineering Department
HITEC University Taxila Cantt, Pakistan

Organization of Lecture

1. DES Variations
   i. Double DES
   ii. Triple DES
      a. Triple DES with three keys
      b. Triple DES with two keys

1. DES Variations (1/7)

→ With the advances in computer hardware (processing speeds, high memory availability); DES is susceptible to possible attacks

↳ Two variations of DES, which are stronger than DES

* Double DES
* Triple DES

1. DES Variations (2/7)

i Double DES

→ does what DES normally does twice

→ Uses two keys say K1 and K2

→ encrypts using K1

→ again encrypts the encrypted text using a different key K2.

→ output is original plaintext encrypted twice.

Encryption

$$\text{P.T} \rightarrow \underset{\uparrow\, K1}{\boxed{\text{encrypt}}} \rightarrow \text{C.T1} \rightarrow \underset{\uparrow\, K2}{\boxed{\text{encrypt}}} \rightarrow \text{C.T2}$$

Decryption

$$\text{C.T2} \rightarrow \underset{\uparrow\, K2}{\boxed{\text{decrypt}}} \rightarrow \text{C.T1} \rightarrow \underset{\uparrow\, K1}{\boxed{\text{decrypt}}} \rightarrow \text{P.T}$$

1. DES Variations (3/7)

⇒ if use 1 bit key two possibilities $\begin{cases} 0 \\ 1 \end{cases}$

" " 2 " " 4 " 00, 01, 10, 11

" " n " " 2^n "

" " 2 different keys : 2^{2n} attempts to crack.

→ Single DES : 2^{56} key search

→ Double DES : $2^{2 \times 56} = 2^{112}$ key search; stronger.

1. DES Variations (4/7)

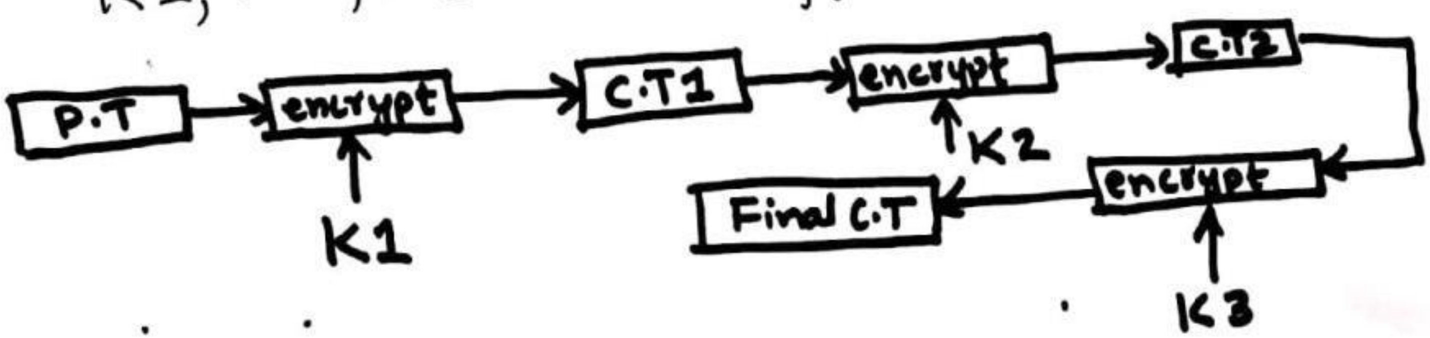
ii Triple DES

→ DES – 3 times
↳ 2 variants of triple DES
- * Use 3 keys
- * Use 2 keys

(a) Triple DES with 3 Keys:

* Plain text block encrypted with K1; then
* Encrypted with second key K2; then
* Encrypted with third key K3

K1, K2, K3 are all different from each other



1. DES Variations (5/7)

encryption:

$$C = E_{K3}\{E_{K2}[E_{K1}(P)]\}$$

$$E_{K2}(C.T1)$$

$$E_{K3}(C.T2)$$

Final C.T

decryption:

$$P = D_{K3}\{D_{K2}[D_{K1}(C)]\}$$

1. DES Variations (6/7)

b) Triple DES with two keys:

→ Triple DES with 3 keys is highly secure but has drawback that requires:

$* \ 56 \times 3 = 168$ bits for key

→ can have a variation with 2 keys

Step 1: Encrypt P.T using K1 ⇒ $E_{K1}(P) \Rightarrow C.T_1$

Step 2: Decrypt output of step 1 with K2:

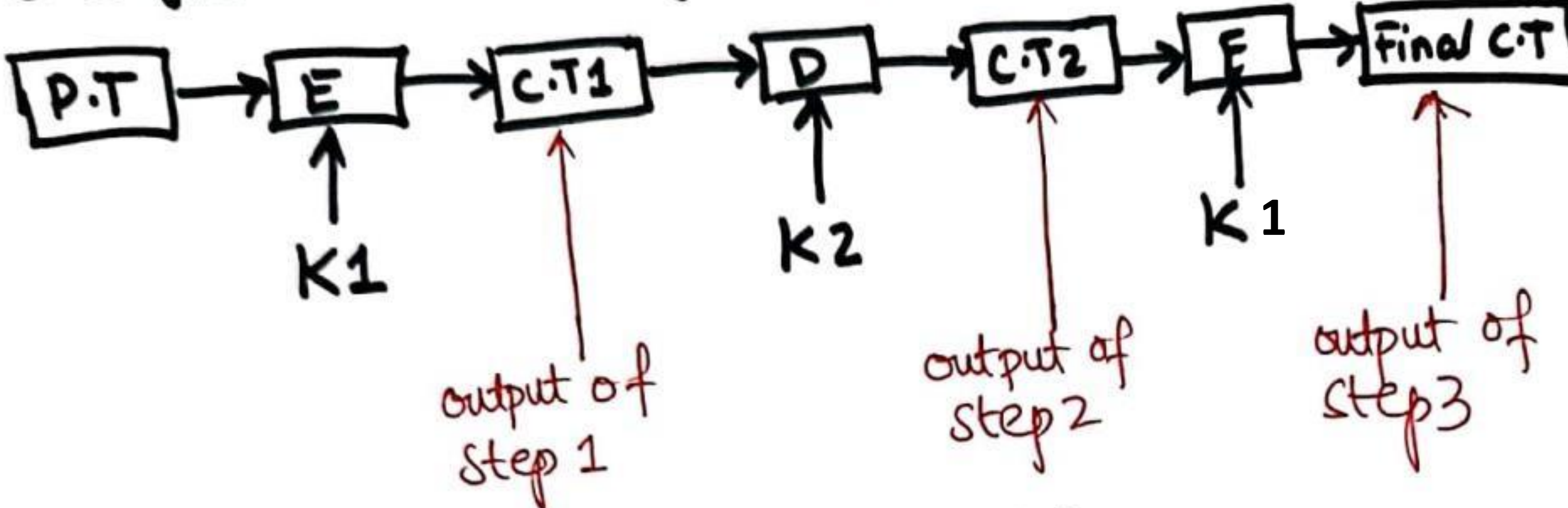
$* \ D_{K2}(E_{K1}(P)) \Rightarrow D_{K2}(C.T1) \Rightarrow C.T_2$

Step 3: Encrypt output of step 2 with K1:

$* \ E_{K1}(C.T2) \Rightarrow$ Final C.T

1. DES Variations (7/7)

encryption $C = E_{K1}\{D_{K2}[E_{K1}(P)]\}$



decryption $P = D_{K1}\{E_{K2}[D_{K1}(C)]\}$

→ Also called Encrypt-Decrypt-Encrypt (EDE) mode.

References: Textbooks

1. *"Cryptography and Network Security", by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2nd Edition, McGraw Hill Education (India) Private Limited.*
2. *"Cryptography and Network Security Principles and Practice", by William stallings, 6th Edition, Pearson.*
3. *"Computer Security Principles and Practice", by William stallings, Lawrie Brown, 4th Edition, Pearson.*